

Computational functionalism, and against intrinsic properties

In response, briefly, to some arguments in The Abstraction Fallacy (Lerchner, 2026)

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Abstract

Alexander Lerchner’s The Abstraction Fallacy seems to be perceived by many as a set of valid arguments against the notion of computational functionalism with regard to phenomenal consciousness. Which it is not. Since a large number of people keep referencing this paper, in the context of this discourse, it may be of use to address some of its points, and to use this to elaborate on some of the notions around computational functionalism and some epistemological/metaphysical ideas around consciousness and its possibility in artificial systems. Especially when it comes to the nature of computation, physics, or the distinction of the real vs. the simulated, metaphysics is a serious and formal discipline. This is not a pick-your-personal-angle practice, especially with regard to the makeup of the entities you use to build your arguments, world-model, or metaphysics. The Abstraction Fallacy is one example of a paper that does not consider its objects with sufficient scrutiny. The main examples of such objects are concepts like intrinsicity, continuous physics, thermodynamic properties, and “the territory”, which Lerchner uses as a basis for his arguments, but which upon closer examination do not actually have the basis he imagines they do.

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1 “Intrinsic physics”

Physics and the entities that populate its theories, such as charges, thermodynamic processes, and the subset of physics that is biology, as well as concepts and categories like causality and time that are used to correlate the entities that populate physics are all nothing more than ways to structure, stabilize, and parse sets of information. The structuring, stabilizing, and parsing of information is the basic necessary epistemic starting point from which any theory, any theoretical entity, and any concept regarding the physical universe or biosphere has been and must be constructed. This is because any possible observation necessarily consists of finitely¹ discernible differences, by definition (this is just what an observation is), as does any thought or concept. The universal principles of this necessary epistemic starting point, that is, of information and difference, have been discovered, and formalized as Turing-universal computation². This means that the notion of “real physics” or “intrinsicity” to charged

¹ Note that we cannot require arbitrary fineness of resolution—infinite resolution is not only not achievable in practice but is in a sense ontologically empty. As Joscha Bach likes to say, “nothing in the universe can depend on knowing the last digit of pi”. Also, as Markus Müller observes, “despite some claims in the opposite, no physical system performing ‘hypercomputation’ has ever been identified” (Müller, 2020, p. 11).

² Once differences are encoded as distinguishable states and their transformations as definite rules, a single universal computational system can simulate any such effective procedure (Turing, 1937, §§ 6–7). Turing universality thus formalizes the general capacity to store, transform, and propagate information.

particles, etc., must be represented and expressed within the set of all possible computational languages, i.e., those representational systems whose atoms are discernibilities.

The idea of a representation-independent “territory” must itself be constructed within the universal principles of representational and formal languages, because no one has ever observed anything but information. Believing that the entities and principles projected into some “reality” subsumed under that term now have properties that exceed the properties of the formal languages used for their construction is a well-known and well-described epistemic misstep, usually referred to as a transcendent metaphysical claim, that is, projecting theoretical entities beyond the theoretical framework that gave birth to them (for Kant’s restriction of the categories to possible experience, see Kant, 1998, A239/B298, A246–247/B303). Assuming the metaphysical or ontic primacy of something specific (in this case, some intrinsic causal physical substrate) is an indication of what would be referred to in analytic philosophy as full-blown pre-critical philosophy—as if Kant had never happened.

Projecting notions that are specific ways of relating information or patterns to each other (e.g., causality or entropy) beyond the domain of all patterns (be they symbolic or sub-symbolic) is usually referred to as hypostasis. It is a type of epistemic misunderstanding where people make claims about ontic primacy or intrinsic properties while forgetting where those notions came from, or how they themselves could ever construct them in the first place. If one is unaware of the principles of construction that give birth to an entity, one risks having no proper epistemic foundations with regard to that entity. We would subsume the broader class of these epistemic misunderstandings under the following taxonomy:

- Naive realism: projecting perceived qualities into objects as intrinsic features. This happens when “the territory” (and whatever it may contain) is spoken of as though it were something that could be accessed and understood independently of its being a model organizing observations, when in fact the entire concept of the territory was constructed within the conceptual apparatus of observations.
- Domain-specific realism: taking the entities of one’s favored discipline to be the furniture of reality itself. This happens when thermodynamic processes, metabolic gradients, and biological constitution are treated as ontically prior to computation, when these are, again, themselves models constructed over observations, stable ways of organizing discernibilities. The “physics of the system itself” is a geometric model that does not somehow stop being a model just because it is familiar.
- Abstraction realism: supposing that because a structure is described at a different level of abstraction, it must therefore be more or less fundamental. People who think of quantum mechanics as more fundamental (in the sense of being closer to “ontic reality”) than molecular physics, which is more fundamental than Newton’s laws, mistake the scale of resolution and the precision to which models can make predictions about structure with closeness to the discernibility-generating substrate. Abstraction depth within a hierarchy of theories is not the same as proximity to any transcendent ontological ground.

All of this happens within the conceptual apparatus. Usually people that go through the argument up until this point go into some branch of idealism or solipsism or ontic anti-realism, but of course the actual epistemic necessity does not take this turn. It derives only that the only theoretical entity one can project beyond the conceptual apparatus is the exact object that can be derived within the space of the conceptual apparatus that meets the criteria for the

conceptual apparatus to be implemented. This is the role we assign to Kant's "thing in itself." In modern terminology, one may refer to it as a Turing-complete-pattern generator³. The thing-in-itself is a boundary-marker, pointed to as the source of discernibilities, but cannot be given any positive characterization, because every positive characterization is itself a structure of differences and so lives inside the apparatus (Kant, 1998, A255/B310–311).

Substrate realism projects a number of entities that were born within the conceptual apparatus onto the level of the thing-in-itself, which is again a transcendent metaphysical statement, the clearest error one can make in epistemology. When substrate realism speaks of "intrinsic, constitutive dynamics" and "the physical territory of experience" (Lerchner, 2026, pp. 5, 9), it takes models, constructions, structures built over differences, and projects them beneath all representation as the ground that representation cannot reach. But the ground in this sense was never anything but representation. We cannot give "the territory" the properties attributed to it (causal power, metabolic constitution, thermodynamic character) beyond what these concepts do to capture particular regularities as patterns in observation, and then say that this "intrinsic beyond" is capable of something that universal pattern generators (computers) are not.

Note that none of this undermines science within its proper domain. Science, much like perception itself, isolates particularly stable observable phenomena and describes these patterns in order to predict future pattern-appearances and make them coherent with other objects, and in some cases imagine the patterns' prior behavior, or developmental trajectory. There are many systems we regularly encounter and have an interest in describing whose evolution is so consistent (or can be approximated to be so) that its description as a sequence of states can be given in a highly compressed form, for instance as some small set of equations. This "computationally reducible"⁴ subset of descriptions of observable systems largely make up what we call "physics". Approximately, some further subset of descriptions within physics, when they have to do with interactions among the patterns we call "atoms", we call chemistry, which contains a further subset of descriptions that, when they have to do with patterns we call "cells"⁵, we call "biology". The names we give to these highly stable computational patterns that we subsume under "physics" (e.g., "atoms", "charges", "fields", "neurons", "voltages") are theoretical state-types whose pointers stabilize in semantics by their being predictive, explanatory, and stable under intervention.

It is in just this way that all computational state-types as models of observables earn their real place in a semantic partitioning into subpatterns of the global pattern of all that can be observed. Any finite set of observations can be described in arbitrarily different, including grotesquely convoluted, ways— fortunately, within the infrastructure of science are

³ This probably also means that the generator is Turing complete (i.e., also a Turing-complete pattern-generator, note the change in hyphenation), but none of this argument depends on making any such characterization, and may only serve to invite criticism that will be much beside the point.

⁴ See Stephen Wolfram's work on computational reducibility (Wolfram, 2002, pp. 737, 749–750).

⁵ Again, approximately: chemistry describes the formation, persistence, and rearrangement of what we call "molecules", which are certain relatively stable combinations of the patterns we call "atoms". Some collections of molecular interactions produce patterns that persist even as their particular constituents are exchanged, and even stabilize bounded, insulated environments in which conditions (which are also patterns) are held consistent. Cells are such organizations. Some such organizations also produce further instances of similar organization, with differences that can be retained through subsequent instances and affect their persistence and further reproduction. Biology describes these processes, their development into more elaborate organizations, and their interactions with each other and their surroundings.

good heuristics, e.g. parsimony⁶, for adjudicating which descriptions stand. All of this still usefully operates within the domain of the apparent, without imposing upon the metaphysical. Physicists usually know that what they are doing is correlating observations and not making transcendent claims about what they observe. The domain of physics in its application is precisely the domain invariant across all possible observations, and the model-lawful closure of the observable, even though there are still some mathematical notions used within physics that are contested (see debates around finitism). Physicists, inasmuch as they are doing physics and not philosophy, are generally not trying to make statements about what cannot even in principle be observed (though they may project their entities into the thing-in-itself in their own minds).

NB: This view of computation and its relation to reality might be best described as “epistemological computationalism” (Bach & Verdicchio, 2012, § 2). Related to (and some might argue, suggesting or recommending of), but strictly distinct from, this is computationalism as metaphysics, that computationalism and thus an ultimately discrete “substrate of the universe” is the minimal necessary and sufficient characterization of reality as such. The physical universe is already a representation, one possible encoding of the observed data. Its success in describing reality has led to a widespread forgetting that it is nonetheless a model fit to observables, but it does not entitle us to treat physical entities as metaphysically primary while regarding every other formal system as merely a theory *about* them. Note that continuity as fundamental to reality is neither observationally nor constructively given, but is an extrapolation of a limit direction within representations. Even if such limits organize our models, it does not follow that they characterize the thing-in-itself. A finite representation of a continuum does not exhibit its completed extension either within the representing mind or as an ontic possibility. Every theory, including a continuous one, must be articulated through discernible states and transformations; every metaphysics necessarily presupposes discernible difference. Universal computation supplies the maximally general framework governing how differences can compose and transform, and it is sufficient to encompass every formalizable physical theory, including continuum models. Treating the physical continuum as the intrinsic constitution of reality therefore adds extra unnecessary apparatus, an ontological posit without adding explanatory or observational capacity. Assuming we care about parsimony in our theories, computationalism is preferable because it retains only the structure every theory already requires while possessing sufficient expressive and generative power for every observable world. Since continuity can neither be itself observed nor logically constructed (and is in fact fairly problematic for our systems of logical construction—again see the finitism debates) and is also not necessary to metaphysically assume to have the appearance of continuity, we should dispense with it as a metaphysical tenet—at least inasmuch as we have widely accepted parsimony as an important heuristic for electing theories. Note that we can also distinguish this from claims about the utility⁷ or convenience of assuming continuity of

⁶ It’s worth noting that parsimony, upon closer examination, goes beyond a useful heuristic and turns out to be a formal necessity of the epistemic process with regard to data and how they inform models. Parsimony as a necessity is an artifact that falls out of the formality of parsing any kind of information, not an assumption that one simply makes with regard to interpreting the world. Parsimony is often misconstrued as a transcendent claim, which it is not. For a related formal account of simplicity and compressibility within an algorithmic-probability framework, see Müller (2020, § 4.1, pp. 12–13).

⁷ Note that contradiction is a stronger notion than utility—things need to be consistent first before they are useful.

certain objects or functions/properties thereof, particularly given the arsenal of mathematical tools that have been built assuming and around notions of continuity.

As discussed earlier, computation in general specifies possible state-types and transformations among them. The limit of resolution fineness is the limit of discernibility, but in many cases it is useful to consider collections of states and transitions as being within equivalence classes. That is, many systems we regularly encounter stabilize state-space partitions that are much more granular than the absolute or even practically available limit of discernibility. The resolution of models of observable state transitions into coarser or finer equivalencing grains often distinguishes particular scientific fields from others (e.g., subatomic physics from chemistry from molecular biology, etc), and is determined by the explanatory goals with respect to which distinctions between states may be considered relevant or not. To observe appearances in an experience of reality, and especially to seek explanation, is to target questions at patterns in experience, which questions each imply a grain at which its proper objects can be distinguished. Science makes an explicit discipline of this, but perception itself does this, segmenting an observable reality into distinguishable components that can be identified as sub-systems, or “functionally equivalenced” with respect to some larger pattern of interest and identification. The patterns of state transition that are available for observation are precisely those that stabilize over micro-state fluctuations, that do not dissipate under perturbation. The existence of macroscopic observables and observers is precisely the result of the functional equivalence of collections of micro-states with respect to the transitions they induce. The natural objects, processes, and properties of observable reality do not themselves have an arbitrarily high-resolution view of the world.

2 Computational functionalism

The observation that phenomena are understood and objects identified in terms of the stabilized equivalence classes of states to which the object/phenomenon is differentially sensitive is called functionalism. A functional model is one that describes a system in terms of transitions between macro-states that correspond to equivalence classes of micro-states that are indistinguishable with respect to the target explanandum. Certain very stable transitions between states or sets of states that remain stable under various forms of perturbation are called causal⁸. Functional models can also be said to describe causal organization. Note that all proper scientific models are functional. The acceptance of particular functional models as scientific theories or common-sense understandings depends on their consistency and thoroughness in describing the stable distinguishable sequences of (equivalence classes of) states. The methodology of science as a way to develop functional models of phenomena generally involves systematically perturbing systems across attractor state boundaries to both determine appropriate state-space partitions and catalogue stable transition sequences. The

⁸ We will not undertake a discussion of causality here, but simply want to point out that causality is a property ascribed to certain trajectories of stable patterns and their interactions, and is projected into certain sets of observations without being part of the observations. Hume argues that we observe succession without directly perceiving a necessary connection between events (Hume, 1902, § VII, part II, pp. 73–76); Kant treats causal connection as a way of parsing information that is prerequisite for experience being as it is, but that is not itself given in perception (Kant, 1998, B233–234).

principle of Occam's Razor is to suggest a preference for partitioning state spaces in as few partitions as possible while still capturing the relevant⁹ phenomena.

Particular proposed functional models may be rejected as good descriptions of observables, but there is no coherent way to be anti-functionalism, or speak of non-functional properties or entities (like intrinsic essences). Any property or entity that participates in observable¹⁰ sequences of state transitions, that has causal effects, can be characterized functionally, as precisely the specification of the effect of its presence or absence on an ambient system's behavior or state evolution. The notion of "input-output mapping" attached to functionalism is often misunderstood (or deliberately misinterpreted, for the sake of argument, drawing on examples of extreme coarse-graining) to suggest naive reductionism or strict behavioralism, but is actually a general way of specifying how systems' states depend on and respond to each other. Properties and entities that are not functional in this way, that cannot even in principle be characterized in terms of their participation in any "mapping" of one set of states to another, that are causally inert with respect to state transitions (how systems evolve or persist over time, prerequisite for its observability at all), are not, and cannot be, part of the modeling of observables. Not only that, such entities are structureless themselves, so not only invisible to the modeling of observables but in a sense invisible even to themselves, and evading characterization or characterizability altogether. Computational functionalism in this sense holds for all phenomena, not just consciousness.

3 Conscious experience

That conscious experience is the necessary epistemic starting point of understanding, and stabilizes and curates the appearances we model in the first place, does not exempt it from being itself among the observables that can be modeled. Conscious experience does seem to be in particular, consistent ways, and its differences make observable differences. That it is such a special phenomenon to so many people is testament to the fact that it indeed has a specific structure distinct from other structures that are not so special to people. The special first-person character of consciousness that distinguishes it from much of the phenomena science concerns itself with is also a functional property¹¹. It may be difficult to provide a comprehensive functional description of consciousness with enough specificity to immediately implement it in an artificial computer, but this difficulty is the general difficulty of science, to determine the coarse-graining that accurately captures the phenomenon and to catalogue the possible transitions and their stabilities. It also may take extra effort to understand what the minimal notion of consciousness is, what is necessary and sufficient versus a particular individual presentation of it, since phenomenal contents of others are not currently possible to observe¹²—but the practice of science is precisely to find the stable abstractions that are sufficient descriptions of systems.

⁹ A distinction between states is relevant to an explanandum when the distinguished states induce (under the interventions in question) different transitions in the explanandum.

¹⁰ Again, by this I do not mean a positivist/behaviorist notion of "what is observable", but the deeper conceptual claim about conceivability/observability as such.

¹¹ Although first-personness and subjectivity do not actually seem necessary for conscious experience (Metzinger, 2024, pp. xiii, 457–459). Here "subjectivity" means an experienced egoic first-person perspective, rather than simply experience occurring in an organism.

¹² That is, phenomenal contents are not currently available for access by a system different from the one constructing the phenomenal contents (or in which phenomenal contents are constructed), in the same representational format in which it is presented to the constructing system.

4 Abstractions vs. the “real world”

The idea that artificial computers cannot be conscious because they merely organize abstract information, and are divorced from some “real physical world”, forgets that human conscious experience is itself the organization of abstract information. The “real physical world” inhabited by humans and from which artificial computers are said to be divorced is a representation, equally proximate (and distant) for immediate contact and knowing by any system organizing available information adequately. Conscious contents are precisely discernible differences (information) stabilized as patterns within experience, whether they are tagged as visual, tactile, remembered, or as abstract thought. Experience is already abstractive, and organizes a variety of stimuli into a unified, immersive representation. The attribution of the sources of information as coming (through vision, for instance) from an external environment, or from within, as bodily sensations or emotional state or abstract thought, is also information, a content of consciousness, functionally characterizable. The self to which contents are attributed is, like everything else knowable, a model over observables, a particular conscious content that can be attended to and interpreted, analyzed, like any other model or collection of observations. That phenomenality and its immediate contents do not present themselves as models (which Thomas Metzinger calls “transparency”¹³) is also a functional quality.

Universal computers are universal pattern generators, capable of realizing any effectively specifiable transition structure and so generate any finitely discernible pattern. Its state-types are attractor basins actively maintained by e.g. noise margins and error correction, forming equivalence classes of states. Differences across class boundaries propagate as the implemented transition function specifies. A computer is a causal insulator¹⁴ for the transition structures it implements. It is a conductor for the set of ambient differences the implemented pattern recognizes as input, amplifying these designated differences into the specified downstream differences. The interpretation of computational implementations as merely abstract mappings labeling the states of suitably partitioned physical systems with possibly arbitrary states of formal machines comes from taking states, instead of transitions, as primary. A state describes what distinctions a system presently maintains, whereas a transition describes how those distinctions constrain subsequent states. To implement is to stabilize particular transition structures with mechanisms that enforce state-types and their transitions against dissipation. The domains of the observable and the implementable coincide for finite observers, whatever can appear for whom can be built at the grain at which it is discernible. Patterns as type, projected beyond finite records of their states, are given only as models that are the result of finite operations of finite observers. We can only project computably, within and in terms of discernibility structures. There can be nothing said (including that it outruns computation) of any “pattern-in-itself” that exceeds every possible record or model thereof. Our reality is the closure of appearance under appearance-operations, e.g. discrimination, composition, projection, negation, all of which are themselves patterns.

¹³ The functional account of phenomenal transparency distinguishes transparent from opaque conscious representations and treats transparency as gradual, rather than necessary for phenomenality (Metzinger, 2003, pp. 356–362). The idea has a predecessor in G. E. Moore’s discussion concerning the difficulty of attending to awareness separately from what one is aware of (Moore, 1903, pp. 446, 450).

¹⁴ Distinct from causal independence. Franz at CIMC likes to capture the distinction by saying computers are “gimbals for languages”, in that a gimbal stabilizes the orientation of, say, a camera, while its support moves, insulating the camera’s orientation from the behavior of the support. This does not make the gimbal causally independent of its support. Likewise, stabilizing a computational organization requires controlled dependence on its implementation, but may causally insulate it from the behavior of its environment.

5 Uncomputable objects

Symbols can be constructed to point at uncomputable objects like the halting set, the completed continuum, or some actualized infinite, but these objects cannot be said to exist or even be “properly conceived” except as limit concepts. These symbols are stand-ins for the rules governing their use as limit objects and as effective procedures for producing other symbols as step-markers of an uncompletable march. The idea that human minds can properly grasp uncomputable or transfinite objects is a hypostasis of the limit, projecting a rule’s unboundedness (the removal of the constraint of specific parameterization, which is to remove information, as abstraction generally does) into an actuality.

To say consciousness is computational is to say that consciousness, like everything else we would like to describe, can only be coherently understood as and in terms of its discernibilities and the discernible ways they unfold. We do not yet have a precise characterization of which class of discernibility structures constitutes consciousness, both because of the ordinary difficulties of science, and because familiarity with and intuitions about consciousness do not require good epistemology or even an interest in understanding consciousness rationally, which has also led to the referent of the term ‘consciousness’ occupying in regular usage a wide distribution over possible concepts. Trying to understand consciousness is also peculiar in that instrument, medium, and object coincide. The intuitions that conscious experience generates of itself are its products, structured and organized alongside other contents of human conscious experience, like models of self and world, and enter the science of consciousness as explananda rather than as constraints. A correct scientific characterization of conscious experience is under no obligation to vindicate or reify the *felt* intrinsicness, unifiedness, or metaphysical primacy of experience, but must predict them, to show why a system so organized would seem to itself in precisely these ways. We cannot deny what seems of conscious experience, which is all that experience is, but can question or reject its self-interpretation as explanatory. This is not to say that experience is ill-defined or confused—conscious experience, phenomenality, is what is here right now, what constructs the present and organizes its contents, and it is in a particular way.

6 Metabolism, and what is actually part of conscious experience

Experience is entirely constituted by its discernibilities; there is nothing that is properly part of experience that makes no difference to experience. The contents of human experience are sensory contents and the world- and self-models built over them that together are a representation of reality. We can theorize about and build analytical models of the “biological substrate” that sustains our consciousness, but the substrate itself as substrate (rather than as models, content) is not part of experience as such, which consists entirely in its content-level organization. We do not, for instance, directly experience neurons’ metabolic processes, though they may be required for sustaining the biological machine on which consciousness runs. Metabolism may be necessary for a self-organizing and -replicating carbon-based substrate to persist in the face of entropic degradation and maintain complex structure long and stably enough to function as a sophisticated information-processing system (which information processing is itself a part of this auto-maintenance), but conscious experience as such is given only as appearance, information, whose totality is the representation exhausting its discernible differences and their mutual relations. To be an aspect of experience is to

be consciously registered, which can only be a property of information organization, how patterns of information maintained across whatever medium enforces them relate to each other. Note that your current experiential state as you are reading this is exactly that as well, a flow of information patterns. Your current visual experience is a changing geometric field of shapes and colors and gradients. There are tingling patterns flowing through your geometric body map. The abstract concepts and associations that are activated as part of comprehending this text are patterns of internal representation. All of these things are wrapped and maintained simultaneously in a representation of perceptions that is itself perceived: second-order perception¹⁵. Consciousness and mental states are not properties of neurons or any “physical” object, but characterize the patterns of communication between many state-holding cells to arrange the signals they have access to into stable representations (usually in ways that are useful for achieving some goal¹⁶).

7 Simulated water

“Simulated water does not make you wet” and analogous arguments distinguish simulation from “true implementation”, and see patterns of information, including conscious experience, that are organized on silicon substrates as somehow less real than (our representation of) the phenomena of “our physics” (for the simulation objection, see Searle, 1980, p. 423; Lerchner, 2026, § 2.5, p. 6). But observing that simulated arithmetic is in fact still arithmetic (and more generally that simulation of computation is still computation), we might conclude that those phenomena that consist in the organization of information, or are representational properties, transcend the “simulation boundary”.

The wetness of water is a model I construct over my experience of water, which is a representation of what I can observe of water. I construct the experience of wetness over the information I have of it as sensory input, and abstractions and inferences thereof. Simulated water does not implement the “wetting” function with respect to me, but simulated water would make a simulated me wet, if the wetting relation is part of the simulation (which presumably it would be, if it were a faithful simulation of water). Simulated me, were I able to construct experience at all¹⁷, constructs the experience of wetness over the information available, just as I do. The “I” that I attach to, and the reality I am embedded in, is also a simulation implemented in a substrate I have no direct access to. We can connect an artificial mind’s internal representations of a world to the world humans construct their

¹⁵ For more on this, see our account of second-order perception (CIMC, 2026, § III.2, pp. 12–13).

¹⁶ It would be possible to recreate mental states and consciousness as information patterns in artificial substrates if we knew exactly how those patterns were structured, without there being explicit goals these states are in service of (though possibly the representation of goals existing would be part of the engineered mental state). In practice, conscious states are likely complex enough that hand-crafting their pattern is implausible as an engineering task (like manually picking the weights in a large neural network), and would be better done by having a system self-organize its information organization in service of some “goal”, setting up a search space of possible system configurations and an efficient way of searching the space for a solution, the way we train artificial neural networks.

¹⁷ Whether simulated me can construct experience at all is, of course, the point in question—but the idea here is that the subjective reality of experience is part of its representation, and is not an extra-representational property that we can attribute to our world but deny to a simulated world from the perspective of a hypothetically conscious artificial entity with similar access to the information of its world. The separate argument that it is reasonable to imagine that an artificial entity could construct experience is given by the general discussion of computational functionalism, that experience is a matter of information organization, and the universality of computation.

internal representations about by attaching sensors to the computer, aligning the content of the artificial mind's world with the one humans approximately share. The representation it would have of that world would be in some sense just as "mere" a representation as we have of our world. Embeddedness in the world humans call real cannot be a definitive constraint on consciousness, since we humans are not "embedded in the real world" either, and only construct a representation of embeddedness from the information we have. The artificial mind also may not need its world to be ours in order for experience to occur, as long as information is available and organized into an experience.

8 Brains in vats, reference, and conclusions on valid entities

We do not yet know if current AI models are conscious, and organize information into experience, but if they do, or some other artificial cognitive architecture could, a lack of connection to the world humans describe with physics may not matter, except to change the content of the artificial world model. Consciousness is not its contents, but what makes its contents possible. All of our understanding of ourselves and the world around us is also mediated by information. We cannot know if we are brains in vats. Because we cannot know, a parsimonious metaphysics would suggest we assume not, since such a statement would make no difference to what we experience as reality. The correct conclusion of Putnam's argument that, were we brains in vats, we could not refer to our containing vats, as the notion cannot refer in experience (for the original argument, see Putnam, 1981, pp. 14–15), is not that we thus must not be brains in vats, but that our knowledge and understanding is only within appearance, and we cannot access a metaphysics of what is "really there", or describe the thing-in-itself. This also means that the conclusion does not need Putnam's very special construction, but is true of all observers, and is the general Kantian idea that we do not have access to anything beyond the realm of appearance and that which situates referents as appearance-patterns within semantics, and gives them meaning. Reference does not need to, and never did, exit the realm of appearance; we have only maps (functions), all the way down. Similarly the sensible reading of the various incompleteness results of mid-20th-century mathematics is not that there is some positively characterizable realm of truth beyond the capture of mathematics and physics, rendering mathematics and physics incomplete, but that such things that are outside the infrastructure of meaning, that gives notions meaning by allocating referent-space in the semantic web we weave with descriptions of reality, cannot be part of our world-model¹⁸.

This is all just to say that if you want to construct arguments and a model of the world using entities you believe to go beyond computation, like continuous physics, or intrinsic properties, the way Lerchner does in *The Abstraction Fallacy*, you need to carefully explain how you were able to know of these objects. Where did you get them from and how did you construct them? If you look carefully at them, you will see that they were computational to begin with, always have been. If you forgo that examination, you will have built your world and ideas upon a collection of square circles.

¹⁸ For more on this, see our Prime Hypostasis discussion (Hildebrandt-Harangozó & Sorensen, 2026).

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